

Human Review: Nearly isometric does not imply almost Lipschitz

Original automatic proof: `solution_packet.pdf`

Checked date: 18 June 2026

Status: Manually checked.

Verdict: The proof appears correct.

Human Comments

The proof appears correct. The construction is a clever exploitation of the main difference between nearly isometric embeddability and almost Lipschitz embeddability: in the nearly isometric definition, the embedding is allowed to depend on the requested compression function ρ . The target space can therefore be made large enough to contain a separate component adapted to each admissible compression.

By contrast, the almost Lipschitz condition is more global. In the present example $X = \mathbb{N}_0$, all nonzero distances are at least 1. Choosing an admissible function φ which is bounded below by a positive constant on $[1, \infty)$ turns an almost Lipschitz embedding into an ordinary bi-Lipschitz embedding of the integer ray. Thus it is enough to construct a target Y which satisfies every individual nearly isometric compression request, but which contains no bi-Lipschitz copy of $\mathbb{N}_0$.

Construction

For each admissible compression ρ , the proof chooses a concave, nondecreasing, unbounded function g_ρ such that

$$\rho(t) \leq g_\rho(t) \leq t, \quad \frac{g_\rho(t)}{t} \rightarrow 0.$$

The ρ -component is a copy of $\mathbb{N}_0$ with metric

$$d_\rho(m, n) = g_\rho(|m - n|).$$

Then Y is the wedge, or bouquet, of all these components at their common basepoint.

This immediately gives nearly isometric embeddability. Given a requested pair (ρ, ω) , map $n \in \mathbb{N}_0$ to n inside the ρ -component. On integer distances $k \geq 1$, one has

$$\rho(k) \leq g_\rho(k) \leq k = \omega(k),$$

so the required inequalities hold.

No Bi-Lipschitz Ray

The obstruction to almost Lipschitz embeddability is that Y contains no bi-Lipschitz copy of the usual ray. Suppose, for contradiction, that

$$F : \mathbb{N}_0 \rightarrow Y$$

is bi-Lipschitz. Let

$$R_n = d_Y(F(n), o)$$

be the distance from the common basepoint. The lower Lipschitz estimate gives linear growth:

$$R_n \geq cn - R_0.$$

On the other hand, consecutive points satisfy

$$d_Y(F(n+1), F(n)) \leq C.$$

If $F(n)$ and $F(n+1)$ were in different bouquet components, their distance would be

$$R_n + R_{n+1},$$

which is eventually larger than C . Hence, after some point, consecutive terms must lie in the same component. Therefore the whole tail of the ray is trapped in a single component Y_{ρ_0} .

Inside this component, distances are measured by

$$g_{\rho_0}(|a_m - a_n|).$$

The upper Lipschitz bound on consecutive points implies that the underlying integer jumps $|a_{n+1} - a_n|$ are uniformly bounded. Thus

$$|a_m - a_n| \leq B|m - n|.$$

But g_{ρ_0} is sublinear, so

$$\frac{g_{\rho_0}(B|m - n|)}{|m - n|} \rightarrow 0.$$

This contradicts the lower linear bound required by a bi-Lipschitz embedding.

Review Outcome

Approve this packet as an apparently correct counterexample to the source problem. The key idea is exactly the dependence of the nearly isometric embedding on ρ : the bouquet can accommodate every requested sublinear compression separately, while the almost Lipschitz condition, in this discrete setting, would force a genuine bi-Lipschitz ray, which the bouquet is designed to exclude.